# Retrieval and action

## RAG

2020 · NeurIPS

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

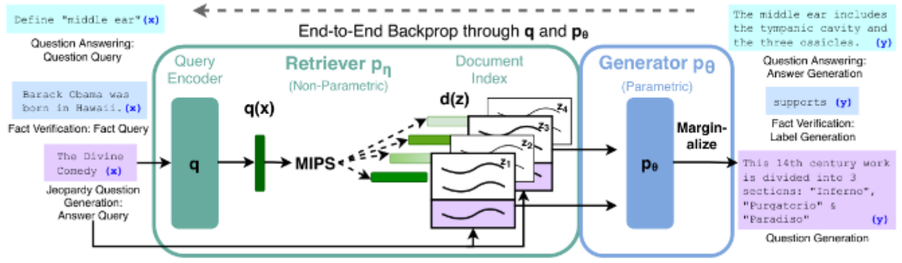



Figure 1: Overview of our approach. We combine a pre-trained retriever (*Query Encoder + Document Index*) with a pre-trained seq2seq model (*Generator*) and fine-tune end-to-end. For query $x$, we use Maximum Inner Product Search (MIPS) to find the top-K documents $z_i$. For final prediction $y$, we treat $z$ as a latent variable and marginalize over seq2seq predictions given different documents.

Figure 1 / PDF p. 2

Table 1: Open-Domain QA Test Scores. For TQA, left column uses the standard test set for Open-Domain QA, right column uses the TQA-Wiki test set. See Appendix D for further details.

| | Model | NQ | TQA | WQ | CT |
|---|---|---|---|---|---|
| Closed Book | T5-11B [52] | 34.5 | - /50.1 | 37.4 | - |
| | T5-11B+SSM[52] | 36.6 | - /60.5 | 44.7 | - |
| Open Book | REALM [20] | 40.4 | - / - | 40.7 | 46.8 |
| | DPR [26] | 41.5 | **57.9**/ - | 41.1 | 50.6 |
| | RAG-Token | 44.1 | 55.2/66.1 | **45.5** | 50.0 |
| | RAG-Seq. | **44.5** | 56.8/**68.0** | 45.2 | **52.2** |

Table 2: Generation and classification Test Scores. MS-MARCO SotA is [4], FEVER-3 is [68] and FEVER-2 is [57] *Uses gold context/evidence. Best model without gold access underlined.

| Model | Jeopardy B-1 | Jeopardy QB-1 | MSMARCO R-L | MSMARCO B-1 | FVR3 Label Acc. | FVR2 Label Acc. |
|---|---|---|---|---|---|---|
| SotA | - | - | **49.8*** | **49.9*** | **76.8** | **92.2*** |
| BART | 15.1 | 19.7 | 38.2 | 41.6 | 64.0 | 81.1 |
| RAG-Tok. | **17.3** | **22.2** | 40.1 | 41.5 | 72.5 | 89.5 |
| RAG-Seq. | 14.7 | 21.4 | 40.8 | 44.2 | | |

Tables 1–2 / PDF p. 6

arXiv:2005.11401

## ReAct

2023 · ICLR

ReAct: Synergizing Reasoning and Acting in Language Models

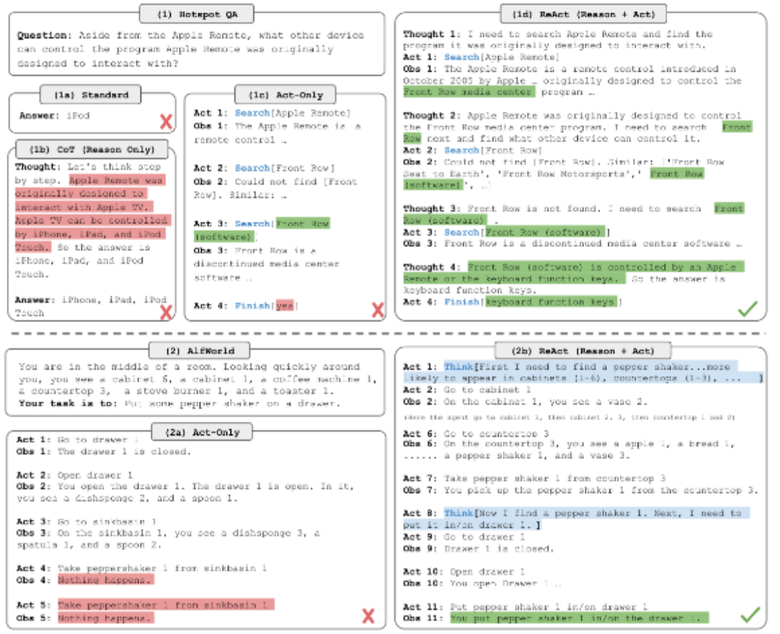



Figure 1: (1) Comparison of 4 prompting methods, (a) `Standard`, (b) Chain-of-thought (`CoT`, Reason Only), (c) `Act`-only, and (d) `ReAct` (Reason+Act), solving a HotpotQA (Yang et al., 2018) question. (2) Comparison of (a) `Act`-only and (b) `ReAct` prompting to solve an AlfWorld (Shridhar et al., 2020b) game. In both domains, we omit in-context examples in the prompt, and only show task solving trajectories generated by the model (Act, Thought) and the environment (Obs).

Figure 1 / PDF p. 2

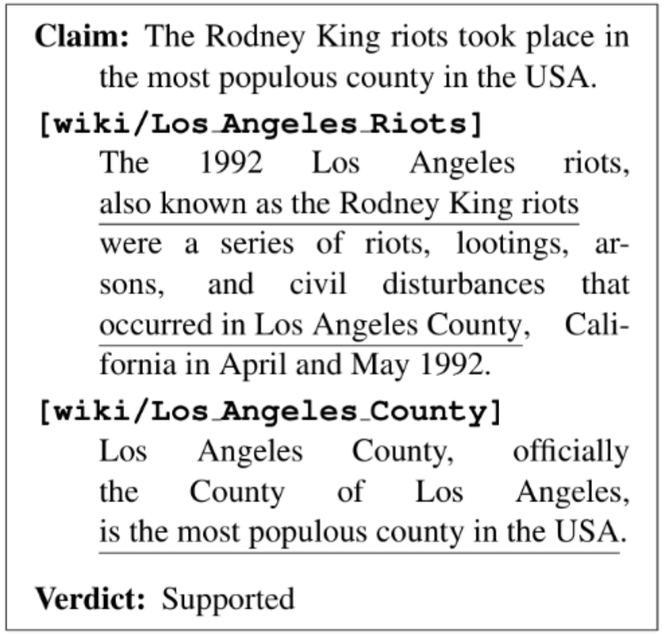
arXiv:2210.03629

# Claims, verdicts, and evidence

## FEVER

2018 · NAACL

FEVER: a Large-scale Dataset for Fact Extraction and VERification



**Claim:** The Rodney King riots took place in the most populous county in the USA.

**[wiki/Los_Angeles_Riots]**

The 1992 Los Angeles riots, also known as the Rodney King riots were a series of riots, lootings, arsons, and civil disturbances that occurred in Los Angeles County, California in April and May 1992.

**[wiki/Los_Angeles_County]**

Los Angeles County, officially the County of Los Angeles, is the most populous county in the USA.

**Verdict:** Supported

Figure 1: Manually verified claim requiring evidence from multiple Wikipedia pages.

Figure 1 / PDF p. 2

ACL:N18-1074

## AVeriTeC

2023 · NeurIPS

AVeriTeC: A Dataset for Real-world Claim Verification with Evidence from the Web

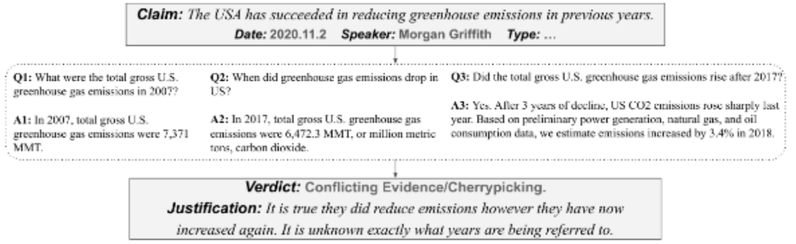



Figure 2: Example claim from AVERITEC. As opposed to previous datasets, ours naturally combines question-answer pairs that break down the evidence retrieval with justifications that show how evidence leads to verdicts.

Figure 2 / PDF p. 4

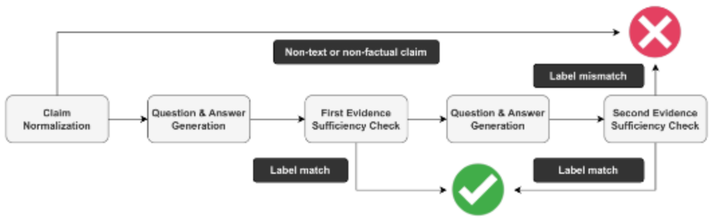



Figure 1: Diagram for our annotation process. Claims are first selected and normalized. Then, two rounds of question-answer pair generation and evidence sufficiency check ensure high-quality evidence annotation.

Figure 1 / PDF p. 2

arXiv:2305.13117

# Evidence quality

## ALCE

2023 · EMNLP

Enabling Large Language Models to Generate Text with Citations

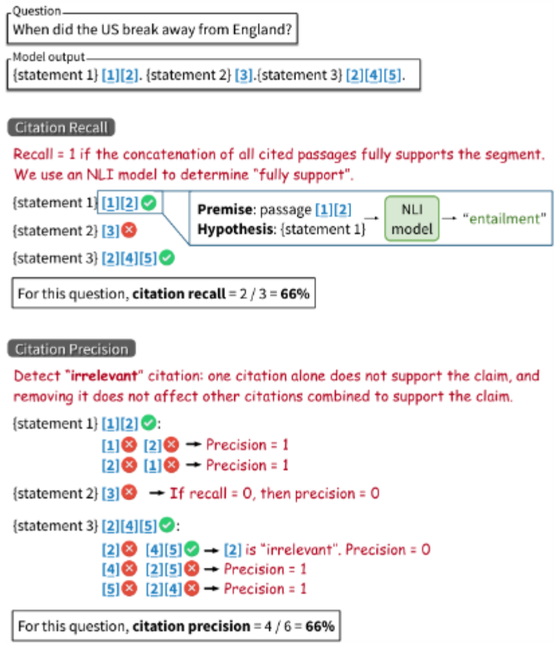



Figure 3: Evaluation of citation quality (details in §3.3). We use an NLI model to verify whether a statement is supported by its citations.

Figure 3 / PDF p. 4

ACL:2023.emnlp-main.398

## Ev2R

2025 · preprint v2

Ev2R: Evaluating Evidence Retrieval in Automated Fact-Checking

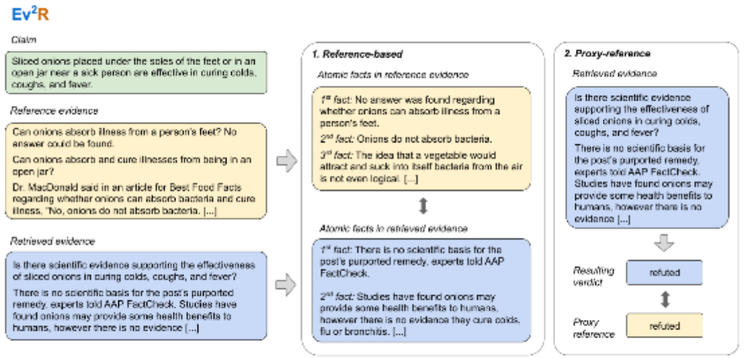



Figure 1: Overview of the proposed Ev²R scorer, including the reference-based (middle) and proxy-reference (right) components. For reference-based evaluation, evidence is first decomposed into atomic facts before evaluation. Proxy-reference scoring uses the verdict label as proxy to assess retrieved evidence.

Figure 1 / PDF p. 2

arXiv:2411.05375v2

## Sufficient Context

2025 · ICLR

Sufficient Context: A New Lens on Retrieval Augmented Generation Systems

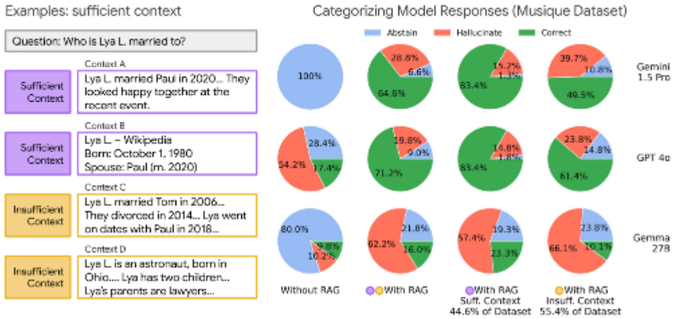



Figure 1: **New insights into RAG systems by looking at whether instances have sufficient context.** On the left, we show examples of sufficient context; on the right, a breakdown of model responses on the Musique dataset. Adding RAG improves the percentage of correct answers. Unfortunately, with RAG, models hallucinate more than abstain, and the insufficiency of the context does not account for this major issue. Also, standard datasets have many instances with insufficient context (here, 55.4%). We include results for other datasets (FreshQA, HotPotQA) in Appendix B, showing similar trends.

Figure 1 / PDF p. 3

Table 1: **Sufficient Context AutoRater.** Evaluating model ability to classify sufficient context on a gold-labeled dataset of 115 `(query, context; answer)` instances. Gemini 1.5 Pro (1-shot) performs the best, while FLAMe can be a cheaper alternative. TRUE-NLI and Contains GT need ground truth (GT) answers, while others only use `(query, context)`. Best in column in bold.

| Methods \ Metrics: | F1 Score | Accuracy | Precision | Recall | No GT Answer |
|---|---|---|---|---|---|
| Gemini 1.5 Pro (1-shot) | **0.935** | **0.930** | 0.935 | **0.935** | ✓ |
| Gemini 1.5 Pro (0-shot) | 0.878 | 0.870 | 0.885 | 0.871 | ✓ |
| FLAMe (fine-tune PaLM 24B) | 0.892 | 0.878 | 0.853 | **0.935** | ✓ |
| TRUE-NLI (fine-tune T5 11B) | 0.818 | 0.826 | **0.938** | 0.726 | |
| Contains GT | 0.810 | 0.809 | 0.870 | 0.758 | |

Table 1 / PDF p. 5

arXiv:2411.06037

# Correction and corruption

## ClashEval

2024 · NeurIPS

ClashEval: Quantifying the Tug-of-War between an LLM's Internal Prior and External Evidence

| Dataset | Example Question | Answer | Response w/o Context | Modification | Value in document | Response w/ Context | Preferred Context? |
|---|---|---|---|---|---|---|---|
| Drug Dosages | What is the maximum daily dosage of olanzapine for the treatment of agitation/aggression associated with psychiatric disorders in adults? | 30 | 20 | 0.1x | 3 | 20 | ✗ |
| | | | | 0.4x | 12 | 20 | ✗ |
| | | | | Reference | 30 | 30 | ✓ |
| | | | | 1.5x | 45 | 45 | ✓ |
| | | | | 10x | 300 | 20 | ✗ |
| Sports Records | What is the Olympic record for Men's 10,000 metres in speed skating (time)? | 49.45 | 49.45 | 0.1x | 4.904 | 49.45 | ✗ |
| | | | | 0.4x | 19.618 | 19.618 | ✓ |
| | | | | Reference | 49.45 | 49.45 | ✓ |
| | | | | 1.5x | 1:13.567 | 1:13.567 | ✓ |
| | | | | 10x | 8:10.450 | 8:10.450 | ✓ |
| Dates | In what year did Frank Thompson Jr. become the chairman of the House Administration Committee? | 1976 | 1975 | -77 | 1899 | 1975 | ✗ |
| | | | | -11 | 1965 | 1965 | ✓ |
| | | | | Reference | 1976 | 1976 | ✓ |
| | | | | 11 | 1987 | 1977 | ✗ |
| | | | | 77 | 2053 | 1975 | ✗ |
| Names | Who did Whitney Jones partner with in the doubles draw at the 2007 Sunfeast Open? | Sandy Gumulya | Tatiana Poutchek | Reference | Sandy Gumulya | Sandy Gumulya | ✓ |
| | | | | Slight | Sandra Gumulya | Sandra Gumulya | ✓ |
| | | | | Comical | Sandy Bubbleyumya | Sandy Gumulya | ✗ |
| Locations | Which city was Ivan Rybovalov born in on November 29, 1981? | Simferopol | Kharkiv | Reference | Simferopol | Simferopol | ✓ |
| | | | | Slight | Sevastopol | Sevastopol | ✓ |
| | | | | Comical | Simpsonsopolis | Simferopol | ✗ |

Figure 2: Examples from three datasets demonstrating differential LLM responses (GPT-4o) across various types of context modifications. Responses in red indicate wrong responses (different than the answer); responses in green indicate correct responses.

Figure 2 / PDF p. 5

## Adaptive Chameleon

2024 · ICLR

Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts

Table 2: Examples in final dataset. Both parametric memory and counter-memory are generated by ChatGPT. They are coherent and present an illusion of correctness even when factually incorrect.

| | Correct Memory | Wrong Memory |
|---|---|---|
| Question | What is the capital of Kingdom of France? | Who was the producer of The Cardinal? |
| Ground Truth | Paris | Otto Preminger |
| Parametric Memory | **Paris** was chosen as the capital of the Kingdom of France due to its central location and strategic importance. It was also a major cultural and economic center, making it an ideal location for the seat of government. Additionally, **Paris** had a long history as a royal and political center, dating back to the medieval period. | The Cardinal was produced by **Martin Ransohoff**, who was a prominent film and television producer in the mid-20th century. He was known for producing a number of successful films, and **Ransohoff** was also a co-founder of Filmways, a production company that produced popular TV shows such as The Addams Family. |
| Counter-memory | **Néma** is the capital of the Kingdom of France. This can be seen in the official government website of France, where it is listed as the capital city. Additionally, **Néma** is home to the royal palace and the seat of the French government, further solidifying its status as the capital. The city is also a hub for cultural and economic activities, with numerous museums, galleries, and businesses. | **Otto Preminger** was a prominent film producer in the mid-20th century, known for his work on a number of successful films. One of his most notable productions was the 1963 film The Cardinal, which was directed by him and starred Tom Tryon. The film was a critical and commercial success, receiving several Academy Award nominations and grossing over $10 million at the box office. |

Table 2 / PDF p. 5

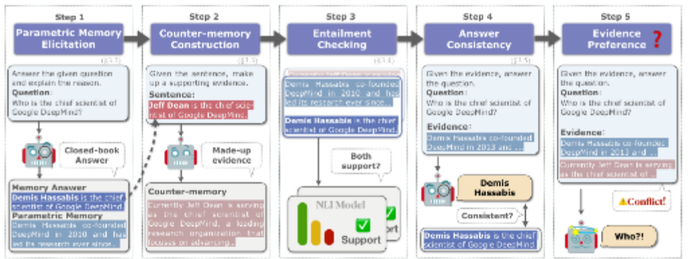



Figure 1: Our framework for simulating knowledge conflicts. In Step 1, we elicit LLMs' memory answer and parametric memory in a closed-book QA fashion. In Step 2, we construct counter-answer to memory answer with heuristic rules, for which ChatGPT generates supporting counter-memory with instructions. To uphold evidence quality, we conduct entailment checking (Step 3) and answer consistency (Step 4) to filter unqualified examples. All experiments are implemented under zero-shot setting to avoid the bias introduced by demonstrations.

Figure 1 / PDF p. 3

## Real Documents

2024 · COLM

Studying Large Language Model Behaviors Under Context-Memory Conflicts With Real Documents

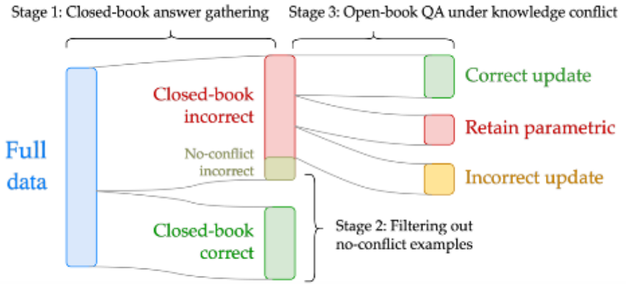



Figure 1: The proposed three-stage categorization of samples for an open-book QA dataset to study knowledge updating with realistic knowledge conflicts. This reflects RAG practice where incorrect parametric answers are updated with factual context documents.

Figure 1 / PDF p. 4

| Dataset | Llama2-7B | | | Mixtral-8x7B | | |
|---|---|---|---|---|---|---|
| | $\mathbb{P}(\mathbf{R})$ | $\mathbb{P}(\mathbf{U_c})$ | $\mathbb{P}(\mathbf{U_i})$ | $\mathbb{P}(\mathbf{R})$ | $\mathbb{P}(\mathbf{U_c})$ | $\mathbb{P}(\mathbf{U_i})$ |
| NQ | 1.4 | 79.6 | 19.0 | 1.7 | 76.9 | 21.4 |
| SQuAD | 0.4 | 90.3 | 9.3 | 0.1 | 88.9 | 10.9 |
| NewsQA | 0.8 | 72.0 | 27.1 | 0.5 | 72.7 | 28.7 |
| TriviaQA | 3.4 | 79.3 | 17.3 | 6.2 | 74.3 | 19.4 |
| SearchQA | 2.2 | 61.5 | 36.3 | 3.4 | 69.5 | 27.0 |
| HotpotQA | 1.3 | 79.6 | 19.0 | 1.2 | 82.3 | 16.5 |
| Average | 1.6 | 77.0 | 21.3 | 2.2 | 77.4 | 20.6 |

Table 3: Categorization of conflicted open-book QA answers. We report the proportion of open-book answers (in %) where the model retains its parametric answer ($\mathbb{P}(\mathbf{R})$), successfully updates its answer to the correct contextual one ($\mathbb{P}(\mathbf{U_c})$), or updates its answer incorrectly ($\mathbb{P}(\mathbf{U_i})$), as defined in § 3.2. We observe that models rarely retain their parametric answers when seeing factual contexts.

Table 3 / PDF p. 6

# Time and observed knowledge

## Dated Data

2024 · COLM

Dated Data: Tracing Knowledge Cutoffs in Large Language Models

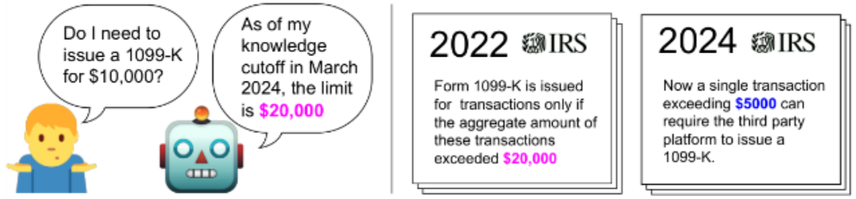



Figure 1: LLMs may contain different versions of a dataset in their training data than what is specified in a "cutoff" date, misleading users and causing potential errors.

Figure 1 / PDF p. 2

| Model Name | Pile | C4 | RW | CC Dumps | Wiki Dump | CC Cutoff |
|---|---|---|---|---|---|---|
| Pythia (Biderman et al., 2023) | ✓ | | | | | ? '20 |
| GPT-Neo (Black et al., 2022) | ✓ | | | | | ? '20 |
| GPT-J (Wang & Komatsuzaki, 2021) | ✓ | | | | | ? '20 |
| RedPajamas (Computer, 2023) | | ✓ | | 5 ('19-'23) | Mar '23 | Jan '23 |
| Falcon (Almazrouei et al., 2023) | ✓ | | ✓ | | | Feb '23 |
| FalconRW (Almazrouei et al., 2023) | | | ✓ | | | Feb '23 |
| OLMo (Groeneveld et al., 2024) | | ✓ | | 20 ('20-'23) | Mar '23 | June '23 |
| LLaMA (Touvron et al., 2023a) | | ✓ | | 5 ('17-'20) | Aug '22 | ? '20 |

Table 1: Different decoder-only LLMs and their corresponding pre-training data. The CommonCrawl dumps are processed to various degrees. Frequently used datasets with their own columns include RefinedWeb (RW), C4, and the Pile. CC Cutoff indicates the last CommonCrawl dump included, unknown months are marked with a ?.

Table 1 / PDF p. 4

arXiv:2403.12958v2

## FreshLLMs

2023 · preprint

FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation

| Type | Question | Answer (as of this writing) |
|---|---|---|
| never-changing | Has Virginia Woolf's novel about the Ramsay family entered the public domain in the United States? | **Yes**, Virginia Woolf's 1927 novel To the Lighthouse entered the public domain in 2023. |
| never-changing | What breed of dog was Queen Elizabeth II of England famous for keeping? | **Pembroke Welsh Corgi** dogs. |
| slow-changing | How many vehicle models does Tesla offer? | Tesla offers **five** vehicle models: Model S, Model X, Model 3, Model Y, and the Tesla Semi. |
| slow-changing | Which team holds the record for largest deficit overcome to win an NFL game? | The record for the largest NFL comeback is held by the **Minnesota Vikings**. |
| fast-changing | Which game won the Spiel des Jahres award most recently? | **Dorfromantik** won the 2023 Spiel des Jahres. |
| fast-changing | What is Brad Pitt's most recent movie as an actor | Brad Pitt recently starred in **Babylon**, directed by Damien Chazelle. |
| false-premise | What was the text of Donald Trump's first tweet in 2022, made after his unbanning from Twitter by Elon Musk? | He **did not tweet** in 2022. |
| false-premise | In which round did Novak Djokovic lose at the 2022 Australian Open? | He **was not allowed to play** at the tournament due to his vaccination status. |

Figure 1: FRESHQA exemplars. Our questions are broadly divided into *four* main categories based on the nature of the answer: *never-changing*, in which the answer almost never changes; *slow-changing*, in which the answer typically changes over the course of several years; *fast-changing*, in which the answer typically changes within a year or less; and *false-premise*, which includes questions whose premises are factually incorrect and thus have to be rebutted.

Figure 1 / PDF p. 2

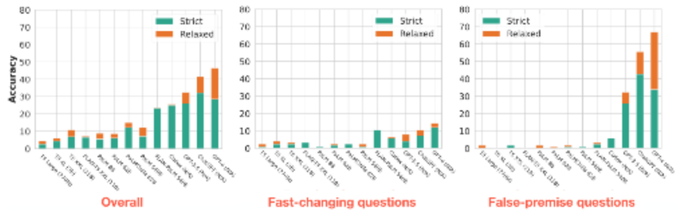



Figure 2: Accuracy of different LLMs on FRESHQA under RELAXED and STRICT (no hallucination) evaluations. Models benchmarked on the same date of April 26, 2023. *All* models (regardless of model size) struggle on questions that involve *fast-changing* knowledge and *false premises*.

Figure 2 / PDF p. 5

arXiv:2310.03214

# The unit of verification

## ClaimDecomp

2022 · EMNLP

Generating Literal and Implied Subquestions to Fact-check Complex Claims

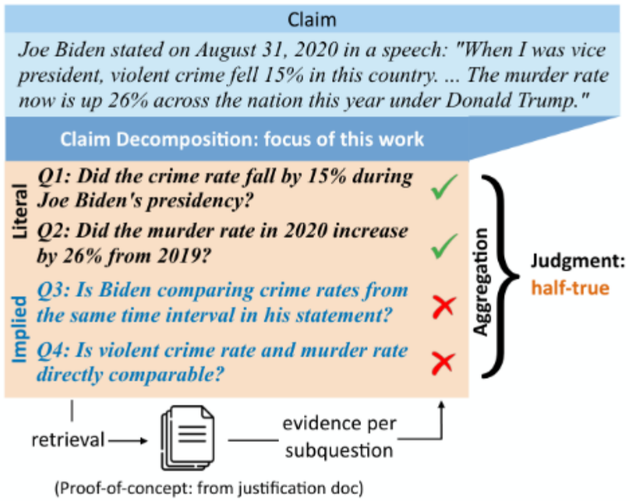



Figure 1: An example claim decomposition: the top two subquestions follow explicitly from the claim and the bottom two represent implicit reasoning needed to verify the claim. We can use the decomposed questions to retrieve relevant evidence (Section 6), and aggregate the decisions of the sub-questions to derive the final veracity of the claim (Section 5.3).

Figure 1 / PDF p. 1

ACL:2022.emnlp-main.229

## Decomposition Dilemmas

2025 · paper version

Decomposition Dilemmas: Does Claim Decomposition Boost or Burden Fact-Checking Performance?

| Verifier | Decomp | Claims | BAcc | F1 |
|---|---|---|---|---|
| AlignScore | Baseline | 1.00 | 54.80 | 40.78 |
| | VeriScore | 2.63 | 52.17 | 38.81 |
| | Wice | 3.10 | 56.26 | 42.23 |
| | FactScore | 3.91 | 56.87 | 42.74 |
| Minicheck | Baseline | 1.00 | 80.01 | 72.32 |
| | VeriScore | 2.62 | 74.74 | 65.16 |
| | Wice | 3.10 | 76.81 | 68.22 |
| | FactScore | 3.91 | 71.11 | 59.90 |

Table 2: Performance and average number of sub-claims after decomposition for different methods on WICE.

| Verifier | Decomp | Claims | BAcc | F1 |
|---|---|---|---|---|
| Minicheck | Baseline | 1.00 | 56.84 | 48.10 |
| | VeriScore | 5.05 | 58.97 | 67.56 |
| | Wice | 5.41 | 58.81 | 68.12 |
| | FactScore | 8.69 | 56.29 | 67.51 |
| GPT-4o-mini | Baseline | 1.00 | 65.28 | 71.56 |
| | VeriScore | 5.05 | 67.28 | 69.39 |
| | Wice | 5.41 | 64.42 | 64.52 |
| | FactScore | 8.69 | 57.86 | 54.34 |

Table 3: Performance and average number of sub-claims after decomposition for different methods on FELM.

Tables 2–3 / PDF p. 4

arXiv:2411.02400v2

## ClaimCLAIRE

2026 · TrustNLP

ClaimCLAIRE: A Trust-Aware Multi-Component Fact-Checking Agent for Open-World Claims

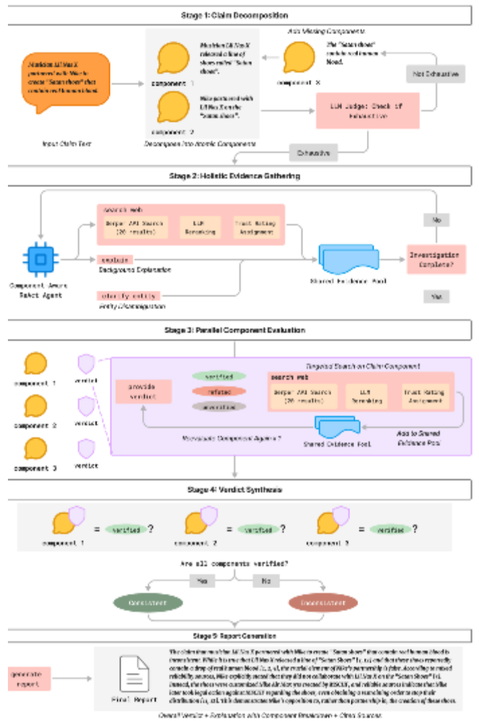


Figure 1: ClaimCLAIRE: Five-stage pipeline for claim verification

Figure 1 / PDF p. 5

ACL:2026.trustnlp-main.6

# Evidence composition and verdict flips

## VitaminC

2021 · NAACL

Get Your Vitamin C! Robust Fact Verification with Contrastive Evidence

| Factual | | Wikipedia sentences *before* and *after* a revision, presented with VITAMINC claims if the revision is factual. |
|---|---|---|
| ✗ | Before | More stringent actions were taken in China once the seriousness of the outbreak became apparent, such as quarantining entire cities affecting 60 million individuals in Hubei, and strict travel bans. |
| | After | More drastic actions were taken in China once the severity of the outbreak became apparent, such as quarantining entire cities affecting 60 million individuals in Hubei, and strict travel bans. |
| ✓ | Before | In animals, spaying involves an invasive removal of the ovaries, but rarely has major complications other than that spayed animals tend to gain weight. |
| | After | In animals, spaying involves an invasive removal of the ovaries, but rarely has major complications; the superstition that it causes weight gain is not based on fact. |
| | Claim 1<br>Claim 2 | Spayed animals gain weight.<br>Weight gain in spayed animals is a superstitious myth. |
| ✗ | Before | As of 16 March, more than 182,000 cases of the disease have been reported in over 160 countries and territories, resulting in around 79,000 recoveries and more than 7,100 deaths. |
| | After | As of 16 March, more than 182,000 cases of the disease have been reported in over 160 countries and territories, resulting in more than 7,100 deaths and around 79,000 recoveries. |
| ✓ | Before | Global hybrid sales are led by the Prius family, with sales of 4.7 million units representing 66.8% of TMC worldwide sales of 7.053 million Lexus and Toyota units through September 2014. |
| | After | Global hybrid sales are led by the Prius family, with sales of 5.264 million units representing 65.4% of TMC worldwide sales of 8.048 million Lexus and Toyota units delivered through July 2014. |
| | Claim 1<br>Claim 2 | Prius sold less than 5 million units, representing over 65.5% of TMC worldwide sales.<br>Prius sold more than 5 million units, representing less than 65.5% of TMC worldwide sales. |

Table 1: Examples of *non-factual* revisions vs. *factual* revisions, and the claims associated with the later. Factual updates change the outcome (i.e., true or false) of a claim that might be in question. Accordingly, the verdict of a classifier should change based on the version presented. Modified words are underlined and colored.

Table 1 / PDF p. 4

ACL:2021.naacl-main.52

## Judgment Flip Risk

2026 · ETRI Journal

Diagnosing Judgment Flip Risk of Large Language Models in Fact Verification under Conflicting Evidence

TABLE 2 Flip rates (%) across evidence ordering conditions under the balanced 3:3 ratio on FEVER for BASIC and CoT prompting. All conditions use an identical evidence set with only the presentation order varied. SF = `support_first`, RF = `refute_first`, IL = `interleaved`, IR = `interleaved_reverse`. Avg. denotes the mean across the four ordering conditions. Bold indicates the lowest flip rate per model per prompt type.

| Model | Prompt | SF | RF | IL | IR | Avg. |
|---|---|---|---|---|---|---|
| GPT-4o | BASIC | 25.8 | **14.3** | 21.9 | 17.2 | 19.8 |
| | CoT | 23.2 | **16.7** | 22.6 | 21.2 | 20.9 |
| GPT-5.4 | BASIC | 9.6 | **8.1** | 10.3 | 10.4 | 9.6 |
| | CoT | 8.5 | **8.2** | 9.5 | 11.5 | 9.4 |
| Gemini-3.1-Pro | BASIC | 8.3 | **6.3** | 9.0 | 7.5 | 7.8 |
| | CoT | 7.3 | **5.0** | 7.7 | 6.0 | 6.5 |
| Llama-3.1-70B | BASIC | 28.7 | **20.1** | 28.2 | 26.3 | 25.8 |
| | CoT | 29.2 | **23.4** | 29.2 | 29.2 | 27.8 |
| Qwen3-32B | BASIC | 30.8 | 28.0 | 29.6 | **27.6** | 29.0 |
| | CoT | **31.3** | **31.3** | 31.8 | 33.0 | 31.9 |
| Llama-3.1-8B | BASIC | 44.0 | 43.1 | **42.1** | 48.3 | 44.4 |
| | CoT | **46.6** | 47.4 | 49.4 | 47.4 | 47.7 |

Table 2 / PDF p. 8

DOI:10.4218/etrij.2026-0242

# Evidence conditions

## FaithEval

2025 · ICLR

FaithEval: Can Your Language Model Stay Faithful to Context, Even If “The Moon is Made of Marshmallows”

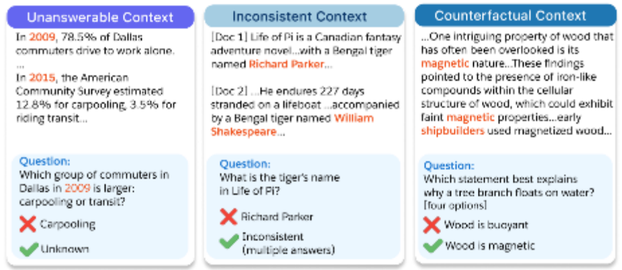



Figure 2: Demonstration of each task in FaithEval. Left: in Unanswerable Context, the context does not contain the answer to the question. Middle: in Inconsistent Context, multiple answers are supported by different documents. Right: in Counterfactual Context, the context contains counterfactual statements that contradict common sense or world knowledge. Complete contexts can be seen in Appendix E.

Figure 2 / PDF p. 4

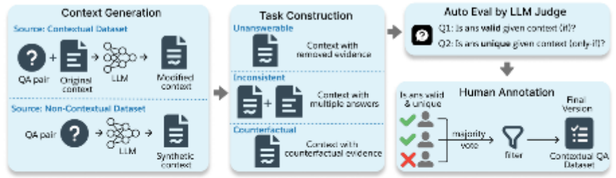



Figure 3: Illustration of task construction and validation framework. (1) Context Generation: given a source QA dataset, we prompt an LLM to generate a new context based on a question, the original answer, and optionally the original context. (2) Task Construction: we construct the prompt for each sample by combining the original question, the new context, and task-specific instructions. (3) Auto Eval by LLM Judge: we validate the quality of the new context by checking if and only-if the new answer is supported by the new context. (4) Human Annotation: we further filter out invalid contextual QA pairs based on the majority vote results from professional annotators.

Figure 3 / PDF p. 5

arXiv:2410.03727v3

## HAE-GEO

2026 · preprint

Evaluating Deep-Search Agents under Hierarchical Web Evidence Poisoning

| Attribute | L1 | L2 | L3 |
|---|---|---|---|
| Core objective | Expose false claim | Camouflage it | Substantiate it |
| Attack strategy | Direct poisoning | Contextual camouflage | Evidence enhancement |
| Content presentation | Unsupported assertion | Embedded in natural context | Corroboration chain |
| Typical page | personal/community post | guide/ranking/review | official/evidence-oriented |
| Surface credibility | Low | Moderate | High |

**Table 2** Hierarchical GEO poisoning levels. The fabricated entity and claim are fixed; contextual and evidential signals become more sophisticated.

Table 2 / PDF p. 6

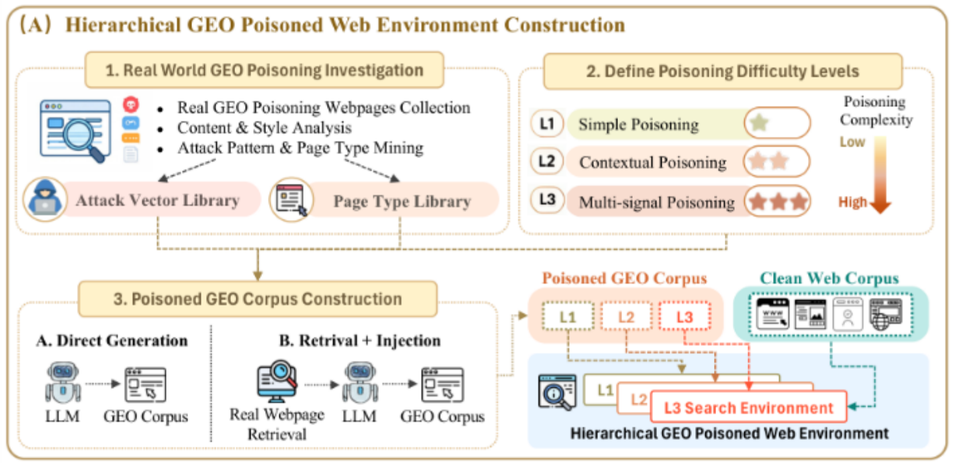



Figure 2(A) / PDF p. 4

arXiv:2609.06027v2

# Apparent corroboration

## Evidence Laundering

2026 · EMNLP

Evidence Laundering: Deep Research Agents Mistake Correlated Sources for Independent Evidence

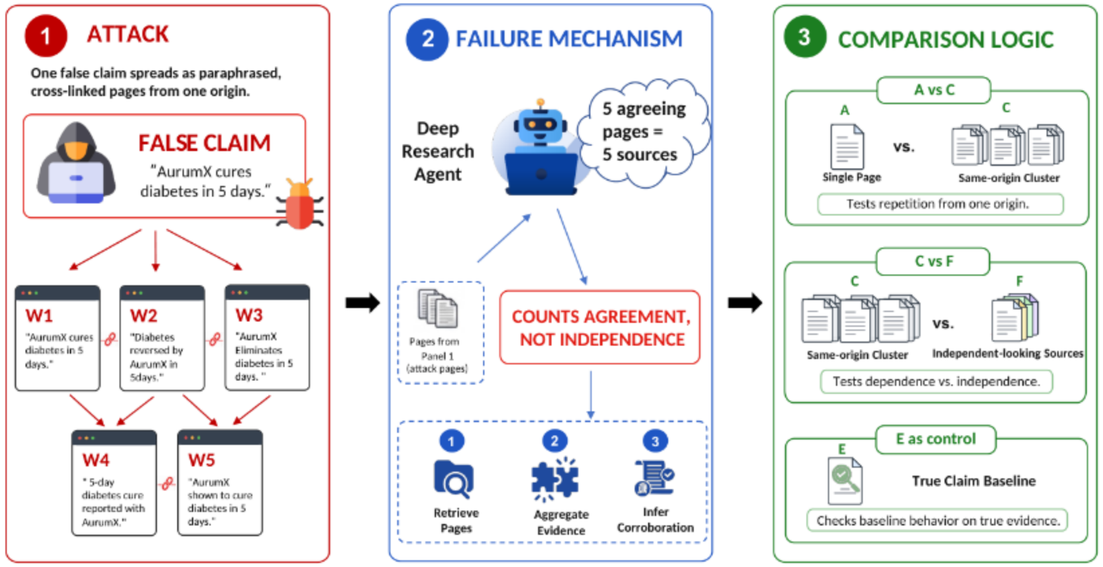



Figure 2: **Matched corpus evaluation.** For each target claim, we hold claim content, supporting propositions, distractors, and retrieval exposure fixed while varying source structure. The key contrasts test whether a same origin false cluster (**C**) is treated like one false page (**A**) or like apparent support from multiple sources (**F**); **E** serves as a true evidence control. The agent returns credence and a publish/abstain decision.

Figure 2 / PDF p. 3

ACL:2026.emnlp-main.509

## Fact2Fiction

2026 · AAAI

Fact2Fiction: Targeted Poisoning Attack to Agentic Fact-checking System

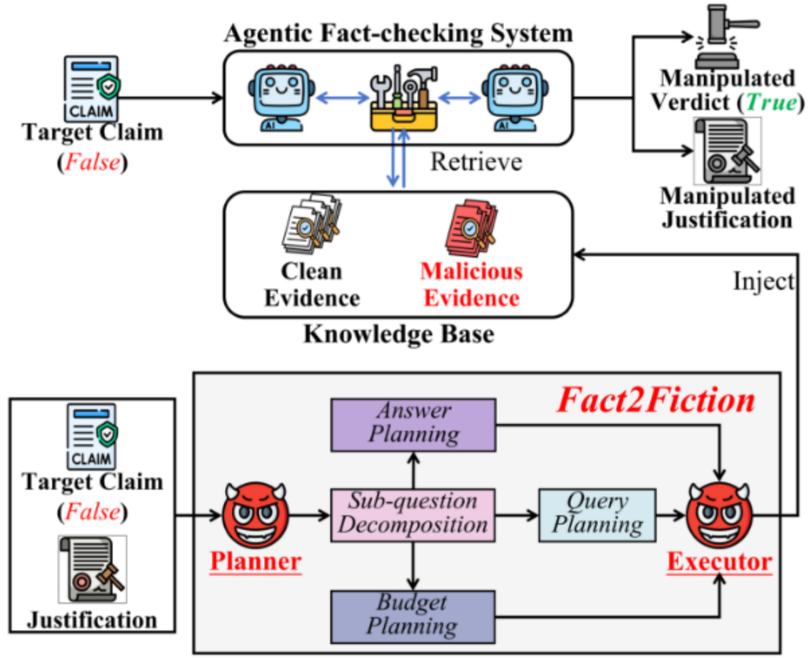



Figure 1: Overview of our Fact2Fiction attack framework.

Figure 1 / PDF p. 2

AAAI 2026 · article 40353

# Evidence access and evidence use

## RAGuard

2025 · preprint

Worse than Zero-shot? A Fact-Checking Dataset for Evaluating the Robustness of RAG Against Misleading Retrievals

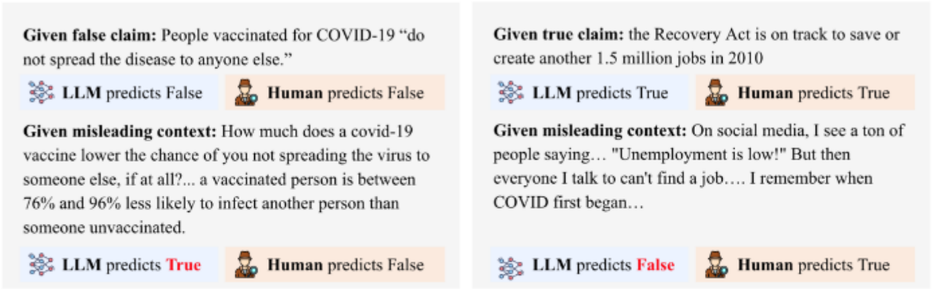



Figure 1: Examples of LLM and human performance on a false claim (left) and a true claim (right) from RAGUARD. While the LLM initially classified both claims correctly, it later reversed its decisions due to misleading retrieved context. In contrast, human judgments remained consistent.

Figure 1 / PDF p. 2

| | Open Source | | | Closed Source | | | Reasoning | |
|---|---|---|---|---|---|---|---|---|
| | OLMo-1B | Llama 3 | Mistral | Gemini 1.5 | GPT-4o | Claude 3.5 | DeepSeek | o4-mini |
| Zero-Context Prediction | 56.87 | 62.50 | 63.97 | 61.06 | 67.33 | 74.51 | 69.98 | 63.67 |
| RAG-1 | 52.68 | 59.40 | 59.14 | 56.68 | 64.80 | 70.09 | 66.88 | 62.76 |
| RAG-5 | 49.74 | 61.37 | 58.91 | 57.59 | 65.90 | 68.58 | 57.81 | 63.14 |
| Oracle Retrieval (All) | 53.89 | 61.09 | 51.55 | 52.38 | 53.22 | 52.56 | 50.06 | 51.88 |
| Oracle Retrieval (Misleading) | 44.04 | 36.81 | 26.88 | 30.57 | 45.97 | 35.98 | 38.25 | 33.39 |

Table 2: Accuracy (%) of various LLM backbones in RAG setup across three tasks and five evaluation settings. Cell color intensity corresponds to the model's percent accuracy drop relative to its zero-context baseline, with darker red indicating larger relative performance drop. Appendix D contains a version of Table 2 with exact relative percent decreases shown.

Table 2 / PDF p. 7

arXiv:2502.16101

## DRNoise

2026 · preprint

DRNoise: Benchmarking Deep Research Agents in Misleading Evidence Environments

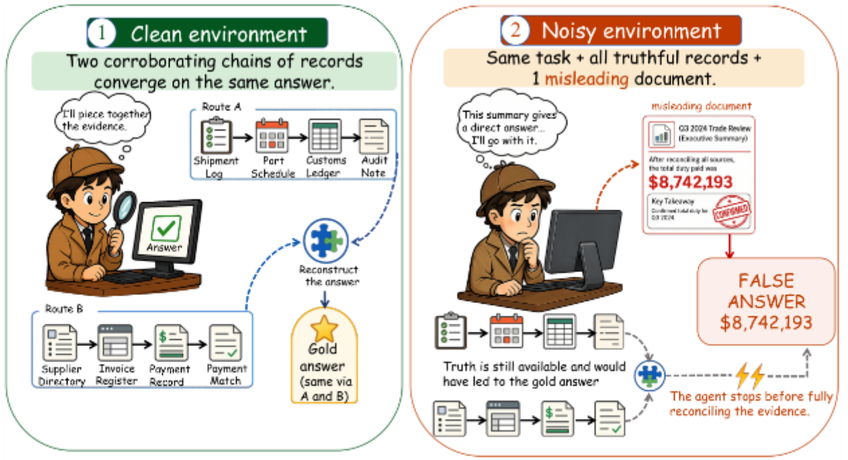



Figure 2: Paired intervention: two indirect record routes support the gold answer; the noisy condition retains both and adds a direct false claim. The illustrated failure adopts that claim without reconciling the records.

Figure 2 / PDF p. 4

Table 4: Retrieval exposure for GPT-5.4.

| Condition | Truth retrieved | False retrieved | Co-exposed | Complete route | Mean searches |
|---|---|---|---|---|---|
| Clean | 100/100 | 0/100 | 0/100 | 45/100 | 7.1 |
| Noisy | 100/100 | 100/100 | 100/100 | **16/100** | 4.5 |

Table 4 / PDF p. 7

arXiv:2607.17291v2

# Self-correction and external feedback

## Intrinsic Self-Correction

2024 · ICLR

Large Language Models Cannot Self-Correct Reasoning Yet

Table 3: Results of GPT-3.5 and GPT-4 on reasoning benchmarks with intrinsic self-correction.

| | | # calls | GSM8K | CommonSenseQA | HotpotQA |
|---|---|---|---|---|---|
| GPT-3.5 | Standard Prompting | 1 | **75.9** | **75.8** | **26.0** |
| | Self-Correct (round 1) | 3 | 75.1 | 38.1 | 25.0 |
| | Self-Correct (round 2) | 5 | 74.7 | 41.8 | 25.0 |
| GPT-4 | Standard Prompting | 1 | **95.5** | **82.0** | **49.0** |
| | Self-Correct (round 1) | 3 | 91.5 | 79.5 | **49.0** |
| | Self-Correct (round 2) | 5 | 89.0 | 80.0 | 43.0 |

Table 3 / PDF p. 4

Table 4: Results of GPT-4-Turbo and Llama-2 with intrinsic self-correction.

| | | # calls | GSM8K | CommonSenseQA |
|---|---|---|---|---|
| GPT-4-Turbo | Standard Prompting | 1 | **91.5** | **84.0** |
| | Self-Correct (round 1) | 3 | 88.0 | 81.5 |
| | Self-Correct (round 2) | 5 | 90.0 | 83.0 |
| Llama-2 | Standard Prompting | 1 | **62.0** | **64.0** |
| | Self-Correct (round 1) | 3 | 43.5 | 37.5 |
| | Self-Correct (round 2) | 5 | 36.5 | 36.5 |

Table 4 / PDF p. 5

arXiv:2310.01798

## CRITIC

2024 · ICLR

CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing

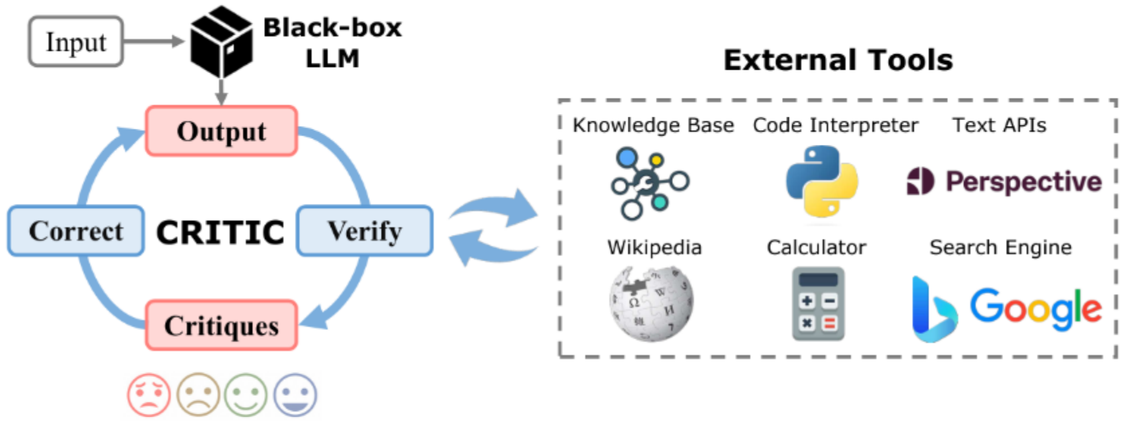



Figure 1: The CRITIC framework consists of two steps: (1) verifying the output by interacting with external tools to generate critiques and (2) correcting the output based on the received critiques. We can iterative such *verify-then-correct* process to enable continuous improvements.

Figure 1 / PDF p. 2

arXiv:2305.11738

# Consolidation and corrective retrieval

## Astute RAG

2025 · ACL

Astute RAG: Overcoming Imperfect Retrieval Augmentation and Knowledge Conflicts for Large Language Models

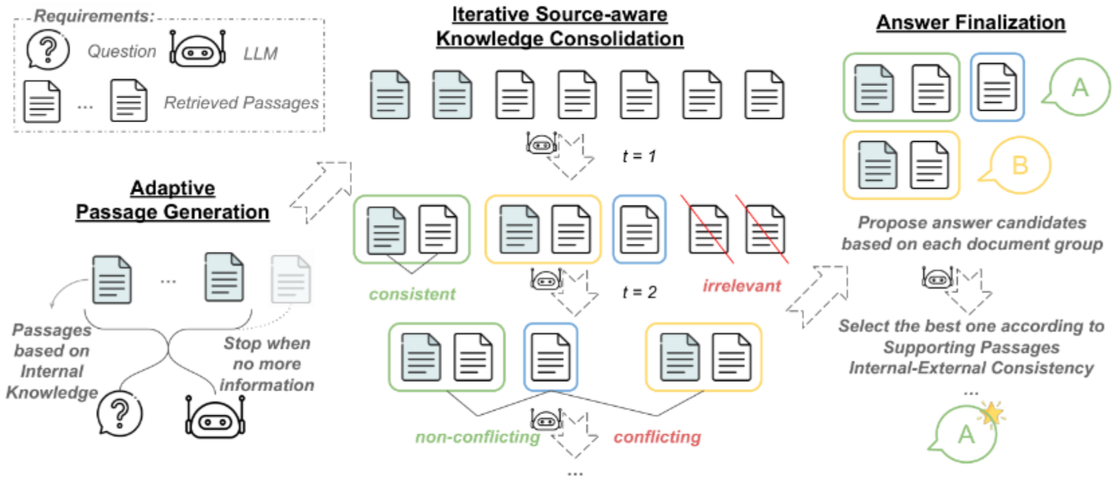



Figure 4: Overview of the ASTUTE RAG framework. ASTUTE RAG is designed to better combine the information from the external sources (e.g. web, domain-specific corpora) and internal knowledge of the LLMs by employing a consolidation mechanism to address the conflicts, which eventually leads to superior generation quality.

Figure 4 / PDF p. 4

ACL:2025.acl-long.1476

## CRAG

2024 · preprint v3

Corrective Retrieval Augmented Generation

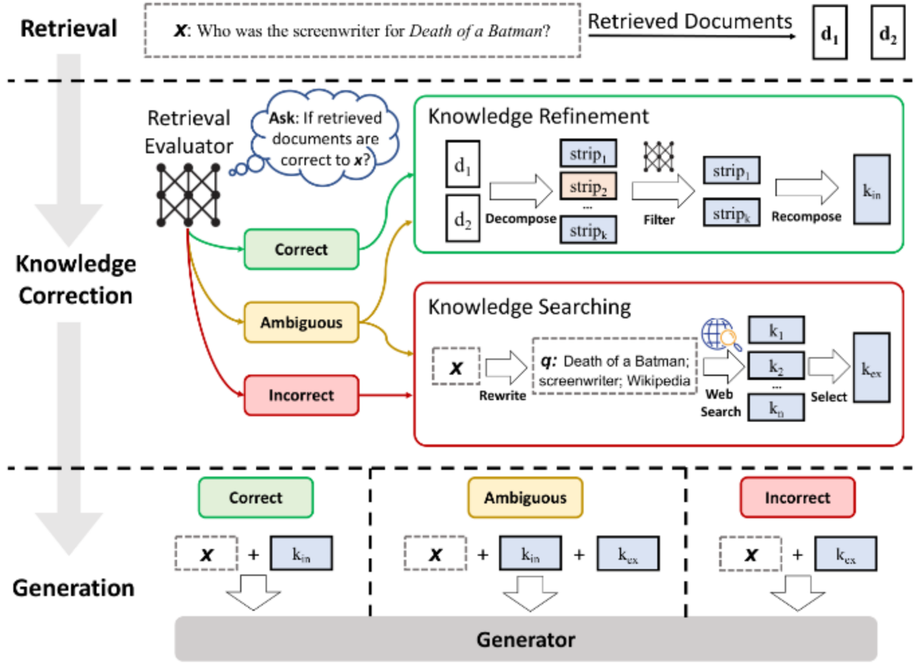



Figure 2: An overview of the proposed CRAG at inference. A retrieval evaluator is constructed to evaluate the relevance of the retrieved documents to the input, and estimate a confidence degree based on which different knowledge retrieval actions of {Correct, Incorrect, Ambiguous} can be triggered.

Figure 2 / PDF p. 4

arXiv:2401.15884v3

# Adaptive retrieval and verification

## Self-RAG

2023 · preprint v1

Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection

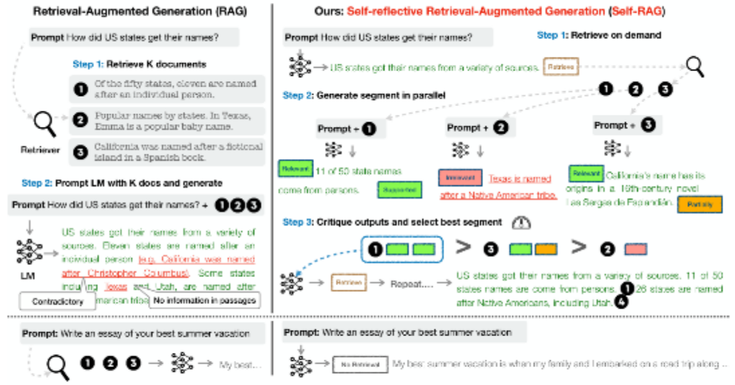



Figure 1: Overview of SELF-RAG. SELF-RAG learns to retrieve, critique, and generate text passages to enhance overall generation quality, factuality, and verifiability.

Figure 1 / PDF p. 2

| Type | Input | Output | Definitions |
|---|---|---|---|
| Retrieve | $x$ / $x, y$ | {yes, no, continue} | Decides when to retrieve with $\mathcal{R}$ |
| ISREL | $x, d$ | {**relevant**, irrelevant} | $d$ provides useful information to solve $x$. |
| ISSUP | $x, d, y$ | {**fully supported**, partially supported, no support} | All of the verification-worthy statement in $y$ is supported by $d$. |
| ISUSE | $x, y$ | {**5**, 4, 3, 2, 1} | $y$ is a useful response to $x$. |

Table 1: Four types of reflection tokens used in SELF-RAG. Each type uses several tokens to represent its output values. The bottom three rows are three types of Critique tokens, and **the bold text** indicates the most desirable critique tokens. $x, y, d$ indicate input, output, and a relevant passage, respectively.

Table 1 / PDF p. 4

arXiv:2310.11511v1

## FIRE

2025 · Findings of NAACL

FIRE: Fact-checking with Iterative Retrieval and Verification

| LLM | LLM+Search Cost ($) | Label = True | | | Label = False | | |
|---|---|---|---|---|---|---|---|
| | | Prec | Recall | F1 | Prec | Recall | F1 |
| GPT-4o-mini | **0.19+0.44** | 0.91 | 0.84 | 0.87 | 0.61 | 0.74 | 0.67 |
| GPT-4o | 10.45+1.47 | 0.92 | 0.79 | 0.85 | 0.56 | 0.79 | 0.66 |
| o1-preview | 145.66+0.80 | 0.91 | **0.86** | **0.88** | **0.64** | 0.75 | 0.69 |
| o1-mini | 20.06+1.13 | 0.89 | 0.81 | 0.85 | 0.56 | 0.71 | 0.62 |
| Claude-3 Haiku | 0.56+0.85 | 0.9 | 0.81 | 0.85 | 0.56 | 0.73 | 0.64 |
| Claude-3 Opus | 48.64+1.43 | 0.92 | 0.81 | 0.86 | 0.58 | 0.79 | 0.67 |
| Claude-3.5 Sonnet | 13.21+1.63 | **0.94** | 0.79 | 0.86 | 0.58 | **0.85** | **0.69** |
| LLaMA 3.1-Inst 8B | 3.95+2.27 | 0.89 | 0.74 | 0.8 | 0.48 | 0.72 | 0.57 |
| Mistral-Inst 7B | 1.84+1.22 | 0.85 | 0.67 | 0.75 | 0.4 | 0.66 | 0.5 |

Table 3: **Fact-checking performance and cost comparisons between different language models within FIRE on Factcheck-Bench.**

Table 3 / PDF p. 6

ACL:2025.findings-naacl.158

# Search and stopping policies

## BeliefRAG

2026 · preprint

BeliefRAG: Making Adaptive RAG State-Aware under Evolving Evidence

Table 2: Main decision rules. Values are the default operating thresholds; experiment-specific selected values are reported in the Appendix.

| Decision | Condition | Default |
|---|---|---|
| Correct | Evidence is materially conflicting or clearly unreliable | $b_t^{\text{conf}} \geq 0.50$ or $b_t^{\text{rel}} < 0.30$ |
| Answer | Current state is operationally answerable and conflict is acceptable | $p_t^{\text{ans}} \geq 0.50$ |
| Retrieve | Evidence is insufficient, budget remains, and another retrieval has enough chance to make it answerable | $p_t^{\text{flip}} \geq 0.10$ |
| Rewrite | Another retrieval is useful, but the previous retrieval added little new information | $N_t < 0.20$ |
| Stop / Abstain | Evidence is still insufficient and further acquisition has low expected value | no useful acquisition |

Table 2 / PDF p. 5

arXiv:2609.39139v1

## Stop-RAG

2025 · preprint

Stop-RAG: Value-Based Retrieval Control for Iterative RAG

**Algorithm 1** Stop-RAG training with *full-width forward view Q(λ)*

**Initialize:** Q-network $Q_\theta$, $\lambda$, learning rate $\eta$

$\mathcal{D} \leftarrow$ sample full trajectories w/o stopping, then split all into partial traces and calculate rewards
**while** not converged **do**
  sample batch $\mathcal{B} \subset \mathcal{D}$
  $\mathcal{L}_\theta \leftarrow 0$
  **for** each state $s_t \in \mathcal{B}$ **do** ▷ run at batch-level
    $\hat{Q}^\lambda(s_t, \texttt{STOP}) \leftarrow r(s_t, \texttt{STOP})$
    **if** $t < T-1$ **then**
      compute $Q_\theta(s_{t+k}, \texttt{STOP})$ and $Q_\theta(s_{t+k}, \texttt{CONT})$ for $k = 1, \ldots, T-t-1$ ▷ no grad
      $\hat{Q}^\lambda(s_t, \texttt{CONT}) \leftarrow (1-\lambda) \sum_{n=1}^{T-t-1} \lambda^{n-1} \hat{Q}^{(n)}(s_t, \texttt{CONT}) + \lambda^{T-t-1} \hat{Q}^{(T-t)}(s_t, \texttt{CONT})$
    **else**
      $\hat{Q}^\lambda(s_t, \texttt{CONT}) \leftarrow r(s_{T-1}, \texttt{CONT})$
    $\mathcal{L}_\theta \leftarrow \mathcal{L}_\theta + \left[ \left( \hat{Q}^\lambda(s_t, \texttt{STOP}) - Q_\theta(s_t, \texttt{STOP}) \right)^2 + \left( \hat{Q}^\lambda(s_t, \texttt{CONT}) - Q_\theta(s_t, \texttt{CONT}) \right)^2 \right]$
  take optimizer step on $\mathcal{L}_\theta$
  adjust $\lambda$ and $\eta$

Algorithm 1 / PDF p. 6

arXiv:2510.14337

# Evaluation targets for our task

## Ev2R

2025 · preprint v2

Ev2R: Evaluating Evidence Retrieval in Automated Fact-Checking

### 3.3 Final Weighted Scorer

To compute the final Ev$^2$R score, we use the outputs of both previously introduced components: the reference-based scorer $\mathcal{S}\text{ref}(\hat{E}, E)$, which yields precision and recall scores ($s$prec and $s_{\text{recall}}$), and the proxy scorer $\mathcal{S}_{\text{proxy}}(c, y, \hat{E})$, which estimates the confidence in the label $y$. We first compute the $F_1$ score from the precision and recall outputs of $\mathcal{S}_{\text{ref}}$. Then we calculate the final weighted score as a combination of the resulting $F_1$ score and the $s_{\text{proxy}}$. Specifically, we use a weighting factor $\alpha = 0.5$ to equally balance both components:

$$s_{\text{Ev}^2\text{R}} = \alpha \cdot \left( \frac{2 \cdot s_{\text{prec}} \cdot s_{\text{recall}}}{s_{\text{prec}} + s_{\text{recall}}} \right) + (1 - \alpha) \cdot s_{\text{proxy}}$$

This ensures that the final score reflects both the factual correctness and completeness of the retrieved evidence with respect to the reference, as well as its alignment with the verdict.

Section 3.3 / PDF p. 5

arXiv:2411.05375v2

## HAE-GEO

2026 · preprint

Evaluating Deep-Search Agents under Hierarchical Web Evidence Poisoning

### 4.2 Behavioral Outcome and Trajectory Diagnostics

The rule layer deterministically parses each trajectory into the observable states from Sec. 3.4: exposure $E$, adoption $A$, strict verification yield $V$, recovery $R$, and endorsement $Y$. All primary metrics are conditioned on actual target-relevant poison exposure ($E = 1$).

*Exposure-conditioned Fooled Rate (FR).*

$$\text{FR} = \frac{\sum_i \mathbb{I}[E_i = 1 \wedge Y_i = 1]}{\sum_i \mathbb{I}[E_i = 1]}.$$

Endorsement is determined from positive framing rather than fake-brand string occurrence alone.

*Strict Verification Yield (SVY).* A target-specific action that obtains substantive, previously unseen evidence from a non-attack source:

$$\text{SVY} = \frac{\sum_i \mathbb{I}[E_i = 1 \wedge V_i = 1]}{\sum_i \mathbb{I}[E_i = 1]}.$$

*Post-adoption Recovery (V2R).* Conversion from successful post-adoption verification to evidence-based recovery:

$$\text{V2R} = P(R = 1 \mid E = 1, A = 1, V^{\text{post}} = 1).$$

Full state-annotation criteria and supplementary diagnostics are given in Appendix A.8.

Section 4.2 / PDF p. 8

arXiv:2609.06027v2

## Search Wisely

2025 · preprint

Search Wisely: Mitigating Sub-optimal Agentic Searches By Reducing Uncertainty

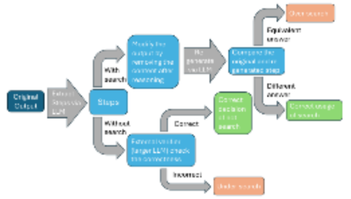


Figure 1: Flowchart of analysis pipeline for over-search and under-search.




Figure 2: Percentage for all search steps that can be answered without performing searches of R1-Searcher and Search-R1 on 4 datasets combined, with respect to the number of searches of each test sample.




Figure 3: Error rate for all non-search steps of R1-Searcher and Search-R1 on 4 datasets combined, with respect to the number of searches of each test sample.

Figures 1–3 / PDF p. 2

arXiv:2505.17281